Null Model and Statistical Testing Framework

KJV 1611 / KJV 1769 Semantic Drift Study

Purpose

This document defines the inferential framework used to test whether observed semantic drift between KJV 1611 and KJV 1769 is lower than expected under randomized conditions.

Core Inferential Objective

The primary inferential question asks whether canonically aligned text unit pairs are significantly more semantically similar than randomized pairings constructed under controlled null conditions. This objective directly supports H1.

Observed Distributions

For each scope and representation, observed distributions are computed from aligned KJV-1611/KJV-1769 pairs. The observed median distance is the primary test statistic because it is robust to upper tail outliers.

Null Model Families

The null procedure holds KJV 1611 source vectors fixed while randomizing KJV 1769 target assignments within a constrained candidate pool. This randomization destroys structural alignment while preserving corpus composition, representation, model, and scope constraints.

Null Model	Scope Constraint	Role
Intratextual	Random candidates drawn from the same work.	Primary inferential baseline
Intracorporal	Random candidates drawn from the same major corpus division.	Secondary robustness check
Supracanonical	Random candidates drawn from the full shared canon.	Exploratory broad corpus check

Permutation Procedure

Each Monte Carlo iteration generates a randomized target assignment for the tested scope, recomputes pairwise distances, and records one randomized median. Repeating this process produces an empirical null distribution of randomized medians.

Number of Permutations

Context	K
Primary (Intratextual)	5,000
Secondary (Intracorporal)	1,000
Exploratory (Supracanonical)	500

Permutation P-Value

The one-sided empirical p-value counts how often randomized medians are as low as or lower than the observed aligned median. An add-one correction prevents zero p-values and reflects finite Monte Carlo resolution.

p = (1 + count(null_median <= observed_median)) / (K + 1)

Secondary Statistical Diagnostics

Mann-Whitney U, Kolmogorov-Smirnov, and Cliff's delta are computed as supporting diagnostics. These tests evaluate stochastic dominance, cumulative distribution separation, and nonparametric effect size but do not replace the permutation median test as the primary inferential basis.

Multiple Testing Policy

Work level tests are adjusted using the Benjamini-Hochberg false discovery rate procedure. Broader intracorporal and supracanonical tests are interpreted as secondary and exploratory robustness evidence.

Study Decisions

Component	Study Decision
Primary null model	Intratextual work level randomization
Secondary null model	Intracorporal work-group randomization
Exploratory null model	Supracanonical full shared canon randomization
Primary statistic	Observed median drift
P-value correction	Add-one Monte Carlo correction
Effect size	SGI
Multiple testing	Benjamini-Hochberg FDR

Expected Output Artifacts

- null_distribution_summary.csv
- statistical_test_results.csv
- baseline_gap_indices_by_scope.csv
- statistical_diagnostics_by_scope.csv